

Fantasy Cricket Fullstack Deployment

Complete Step-by-Step Production Guide: Localhost • Git • MongoDB Atlas • Render • Vercel

▲ Vercel (Frontend)

⚙️ Render (Backend)

👉 MongoDB Atlas (Cloud DB)

1. Monorepo Architecture Overview

The platform is organized into two independent sub-projects managed within a single repository:

Component	Technology Stack	Hosting Platform	Live Production URL
Frontend (UI)	React 19, TanStack Start, Vite, Tailwind CSS	Vercel	<code>https://fantasy-cricket-integrated-qr4b.vercel.app</code>
Backend (API)	Node.js, Express, TypeScript, Mongoose	Render	<code>https://fantasy-cricket-api-ro18.onrender.com</code>
Database	MongoDB Atlas (M0 Shared Cloud Cluster)	MongoDB Cloud	<code>mongodb+srv://.../fantasy_cricket</code>

2. Localhost Setup & Running Locally

Before deploying to cloud servers, run the platform locally:

```
# 1. Install dependencies
cd API && npm install
cd ../UI && npm install
cd ..

# 2. Start Backend (Runs on http://localhost:5000)
npm run dev:api
```

```
# 3. Start Frontend (Runs on http://localhost:8080)
npm run dev:ui
```

3. Git Version Control & Deployment Workflow

3.1 Initial Setup

```
git init
git add .
git commit -m "feat: initial commit"
git branch -M main
git remote add origin https://github.com/Vikas-Sharma-25/Fantasy-Cricket-Integrated.git
git push -u origin main
```

3.2 Continuous Deployment Workflow

Whenever you make any changes locally, run:

```
git add .
git commit -m "fix: updated features"
git push origin main
```

💡 **Automated Cloud Triggers:** Running `git push origin main` automatically informs both **Render** and **Vercel** to pull latest code and trigger zero-downtime production deployments!

4. MongoDB Atlas Setup & Local DB Migration

4.1 Setting up Atlas (Free Cloud Database)

1. Sign in at cloud.mongodb.com.
2. Create an **M0 Free Cluster** in AWS Mumbai / Singapore.
3. Under **Database Access**, create user `vikass78901_db_user` with password `VikasCricket2026`.
4. Under **Network Access**, add IP: `0.0.0.0/0` (Allow access from anywhere).
5. Copy connection string:
`mongodb+srv://vikass78901_db_user:VikasCricket2026@cluster0.oh8z9di.mongodb.net/fantasy_cricket?retryWrites=true&w=majority`

4.2 Migrating Local Data to Atlas

Run the 1-click migration script to copy all existing local collections (Matches, Players, Users, Contests, Teams) to Atlas:

```
cd API
node src/scripts/migrate.js
"mongodb+srv://vikass78901_db_user:VikasCricket2026@cluster0.oh8z9di.mongodb.net/fantasy_cricket?retryWrites=true&w=majority"
```

4.3 MongoDB Issues Encountered & Solutions

✗ Issue 1: "The limit for free tier clusters in this project has been reached"

Cause: MongoDB allows 1 free cluster per project.

Fix: Used the existing active cluster in Project 0 instead of creating duplicates.

✗ Issue 2: "bad auth : authentication failed" during migration

Cause: The placeholder `<db_password>` was passed literally in the URL.

Fix: Replaced with the real user password `VikasCricket2026` without brackets.

5. Backend Deployment on Render

5.1 Service Configuration Table

Setting	Value	Explanation
Name	Fantasy-Cricket-API	Unique name of the web service
Language	Node	Runtime environment
Root Directory	API	Points directly to the backend subdirectory
Build Command	npm install && npm run build	Installs packages and compiles TypeScript into dist/
Start Command	npm start	Executes node dist/server.js

Setting	Value	Explanation
Instance Plan	Free (\$0/month)	512 MB RAM, 0.1 CPU

5.2 Render Environment Variables

```

NODE_ENV=production
PORT=5000
CLIENT_URL=https://fantasy-cricket-integrated-qr4b.vercel.app,http://localhost:8080
MONGO_URI=mongodb+srv://vikass78901_db_user:VikasCricket2026@cluster0.oh8z9di.mongodb.net/fantasy-cricket?retryWrites=true&w=majority
JWT_ACCESS_SECRET=fantasy_access_secret_super_secure_key_2026
JWT_REFRESH_SECRET=fantasy_refresh_secret_super_secure_key_2026
JWT_ACCESS_EXPIRES_IN=2d
JWT_REFRESH_EXPIRES_IN=7d
OTP_LENGTH=6
OTP_EXPIRES_IN_MINUTES=5
OTP_RESEND_COOLDOWN_SECONDS=30
OTP_MAX_ATTEMPTS=5
OTP_MAX_RESEND_PER_WINDOW=5
MAIL_FROM=vikass78901@gmail.com
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=vikass78901@gmail.com
SMTP_PASS=lwvrqtwrvxczfnvh
SMTP_SECURE=false
RATE_LIMIT_WINDOW_MINUTES=15
RATE_LIMIT_MAX=2000

```

5.3 Render Issues Encountered & Solutions

✗ Issue 1: TS5108 / TS5102 (moduleResolution=node10 & baseUrl removed in TypeScript 5.8)

Cause: Render uses modern TypeScript compiler where legacy compiler flags error out.

Fix: Cleaned `API/tsconfig.json` to standard modern ES2020 compilation.

✗ Issue 2: TS7016 "Could not find a declaration file for module 'express'"

Cause: `NODE_ENV=production` causes `npm install` to omit `devDependencies` where `@types/*` resided.

Fix: Moved `@types/express`, `@types/bcryptjs`, `@types/jsonwebtoken`, etc. directly into `dependencies` in `API/package.json`.

✗ Issue 3: MongooseServerSelectionError (IP Not Whitelisted)

Cause: Render servers connect from dynamic cloud IPs.

Fix: Added `0.0.0.0/0` in MongoDB Atlas Network Access.

6. Frontend Deployment on Vercel

6.1 Vercel Configuration

1. Import GitHub repository into Vercel.
2. In Project Settings, set **Root Directory** to `UI`.
3. In **Environment Variables**:
Key: `VITE_API_URL`
Value: `https://fantasy-cricket-api-ro18.onrender.com/api`
Type: **Config** (Public)
4. Trigger Deployment → Status becomes **Ready!**

6.2 Vercel Issues Encountered & Solutions

✗ Issue 1: "404: NOT_FOUND" on initial deployment

Cause: Vercel was building the monorepo root rather than `UI/`, and missing Nitro's Vercel build target.

Fix: Set Root Directory to `UI` and configured `"build": "NITRO_PRESET=vercel vite build"` in `UI/package.json`.

✗ Issue 2: "Environment variables with public prefix cannot use visibility: secret"

Cause: `VITE_` variables are client-facing and cannot be marked as write-only backend secrets.

Fix: Deleted old secret variable and recreated `VITE_API_URL` with **Config** visibility.

7. Production Verification Summary

Endpoint / Feature	Target URL	Verified Result
Frontend UI	<code>https://fantasy-cricket-integrated-qr4b.vercel.app</code>	✅ 200 OK (Full responsive UI)
Backend Health API	<code>https://fantasy-cricket-api-ro18.onrender.com/api/health</code>	✅ <code>{"success": true, "message": "OK"}</code>
Database	MongoDB Atlas Cluster	✅ 100% Local DB synced

 **Deployment Complete!** The Fantasy Cricket Platform is officially live in production.